Geometry
Unit 11
Lesson 1 Practice

Name____________________
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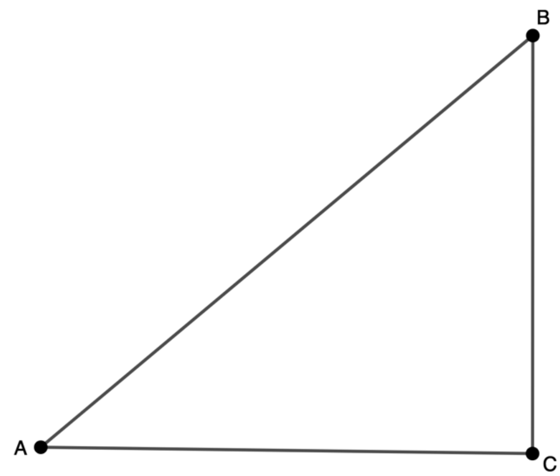
1. Construct the three perpendicular bisectors for ΔABC.



2. Label the circumcenter of the triangle as Point R.

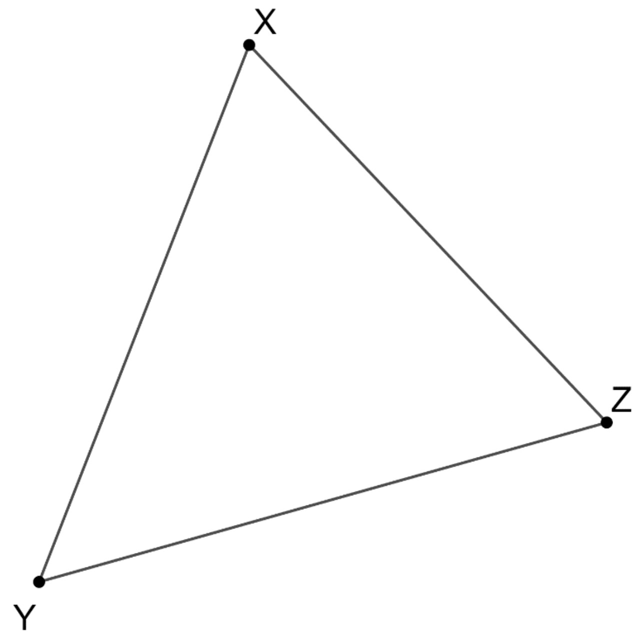
3. Use a ruler to measure the length of $\overline{AR}$.

4. Use a ruler to measure the length of $\overline{CR}$.

5. Use a ruler to measure the length of $\overline{BR}$.

6. Use a compass to draw the circumscribed circle around the triangle.

7. Construct the three perpendicular bisectors for ΔXYZ.



8. Label the circumcenter of the triangle as Point T.

9. Use a compass to draw the circumscribed circle.

10. Given that the measure of $\widehat{XZ} = 100°$, what is the measure of $\angle Y$?

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Lesson 2 Practice

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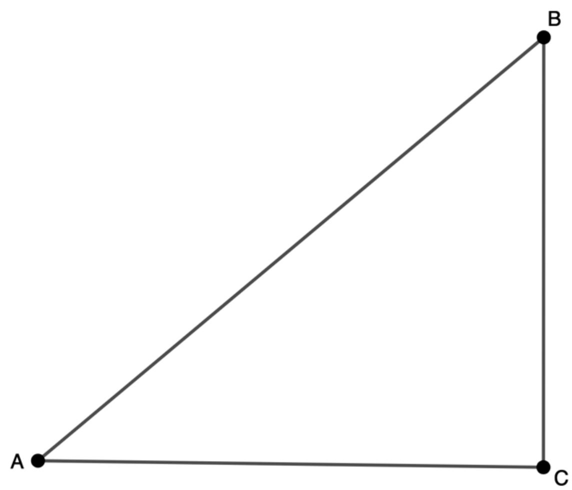
Use ΔABC to answer questions 1-5.

1. Construct the three angle bisectors for ΔABC.



2. Label the incenter of the triangle as Point Z.

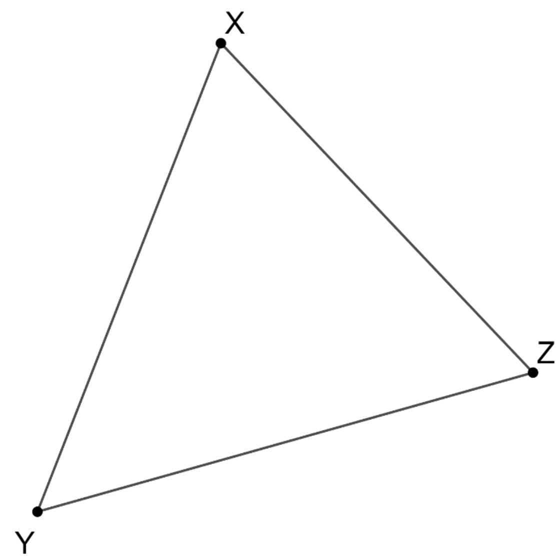
3. Use a compass to draw the inscribed circle inside the triangle, with point Z as its center.

4. Label the three points where the inscribed circle intersects the triangle as F, G, and H.

5. Measure the lengths of segments $\overline{FZ}$, $\overline{GZ}$, and $\overline{HZ}$.

Use ΔXYZ to answer questions 6-10.

6. Construct the three angle bisectors for ΔXYZ.



7. Label the incenter of the triangle as Point K.

8. Use a compass to draw the inscribed circle inside the triangle, with point K as its center.

9. Label the three points where the inscribed circle intersects the triangle as $M, P,$ and T.

10. Measure the lengths of segments $\overline{MK}, \overline{PK}$, and $\overline{TK}$.

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Lesson H1 Practice

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Complete the missing steps showing one way to construct an equilateral triangle inscribed inside a circle.

Step	Procedure
1	Use a compass to draw a circle. Mark the center and label it.
2	
3	Draw a diameter. Label the endpoints.
4	
5	Draw a circle.
6	
7	

Use Point A to construct an equilateral triangle inside a circle.

Complete the missing steps showing one way to construct a square inscribed inside a circle.

Step	Procedure
1	Use a compass to draw a circle. Mark the center and label it.
2	
3	Construct a circle with a radius larger than the original circle with the center at one endpoint of the diameter.
4	
5	Mark the intersections of the two circles.
6	
7	
8	Connect the points.

Use Point P to construct a square inside a circle.

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Lesson 3 Practice

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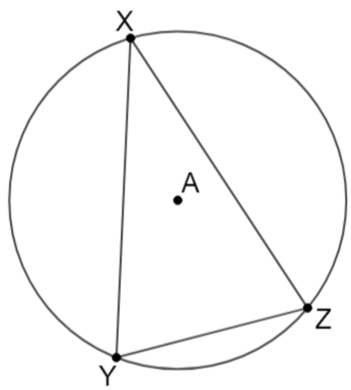
Circle A is shown, where points $X, Y,$ and Z are on Circle A and create ΔXYZ. $m\widehat{XZ} = 140°$, $m\angle ZXY = 30°$, and $m\angle XYZ = (2x - 18)°$.



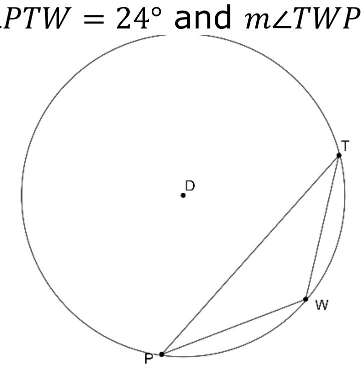
1. Find the value of x.

2. Find the $m\angle XYZ$.

3. Find the $m\widehat{YZ}$.

4. Find $m\angle YZX$.

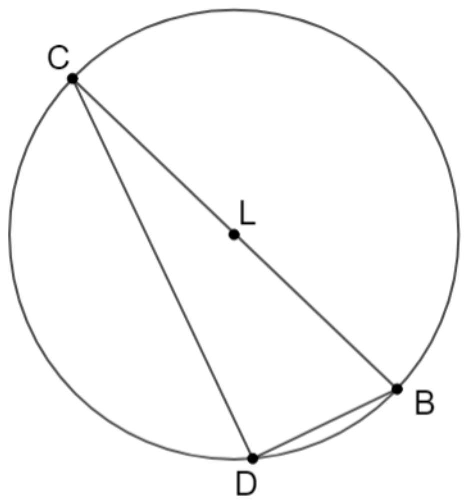
Circle D is shown, where points $P, T,$ and W are on Circle D and create ΔPTW. $m\angle PTW = 24°$ and $m\angle TWP = 140°$.



5. Find $m\widehat{PT}$.

6. Find $m\widehat{TW}$.

Circle L has points $B, C,$ and D on Circle L and diameter $\overline{BC}$. ΔBCD is inscribed in the circle. $m\widehat{CD} = 160°$, $m\angle DCB = 10°$, and $m\angle DBC = (2x + 20)°$.



7. What is the $m\angle CDB$?

8. What is the value of x?

9. Determine $m\angle DBC$.

10. Determine $m\widehat{DBC}$.

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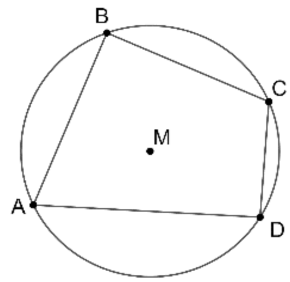
Circle M is shown, with $ABCD$ inscribed in the circle. $m\angle ABC = 96°$, and $m\angle BCD = 125°$.

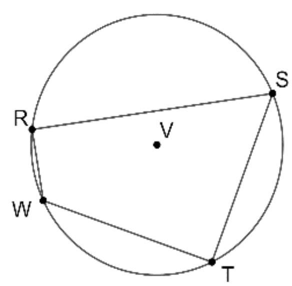


1. Find the $m\angle ADC$.

2. Find the $m\angle DAB$.

3. Find the $m\widehat{ABC}$.

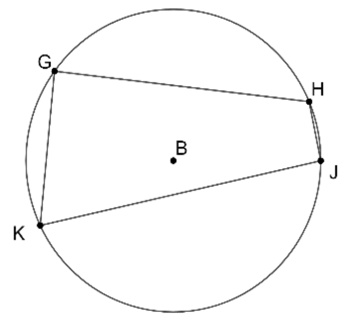
Circle V is shown, with $RSTW$ inscribed in the circle. The $m\angle RWT = 140°$, $m\angle WRS = (5x)°$ and $m\angle RST = (2x + 6)°$.



4. Find the value of x.

5. Find the $m\widehat{WTS}$.

Circle B is shown, with $GHJK$ inscribed in the circle. $m\angle GHJ = (3x + 50)°$, $m\angle HJK = (4x)°$ and $m\angle GKJ = (4x - 10)°$.



6. Find the value of x.

7. Find $m\angle KJH$.

8. Find $m\angle GKJ$.

9. Find $m\angle HGK$.

10. Find $m\angle JHG$.

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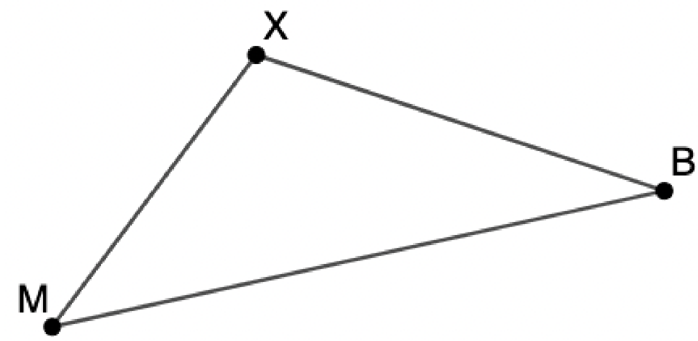
ΔMXB is shown where $m\angle M = (5x)°$, $m\angle X = (15x - 10)°$, and $m\angle B = (5x - 10)°$.



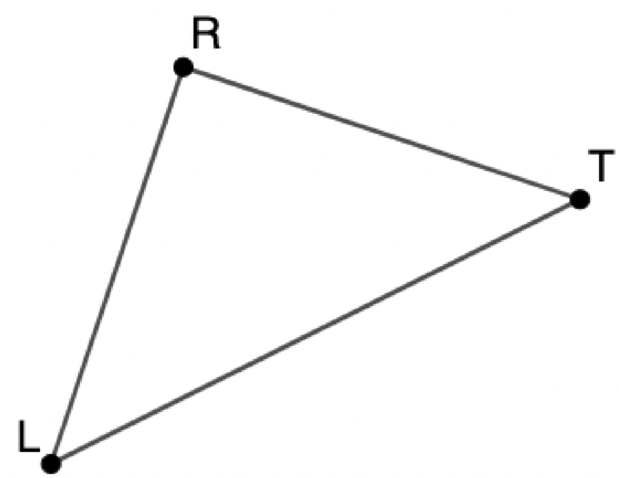
1. Find the value of x.

2. What is the $m\angle M$?

3. What is the $m\angle X$?

4. What is the $m\angle B$?

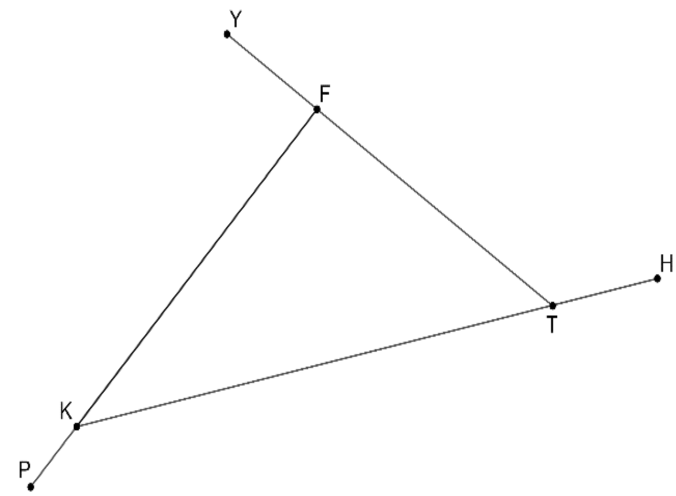
ΔLRT is shown, where $m\angle L = m\angle T = (2x + 6)°$ and $m\angle R = (4x + 12)°$. Figure is not drawn to scale.



5. Find the value of x.

6. What is the $m\angle R$?

ΔFKT is shown, where $\angle KFY$, $\angle PKT$, and $\angle FTH$ are exterior angles with $m\angle FTH = (12x + 25)°$, $m\angle YFK = 95°$ and $m\angle FKT = 24°$.



7. Find the value of x.

8. What is $m\angle KTF$?

9. What is $m\angle PKT$?

10. What is $m\angle KFT$?

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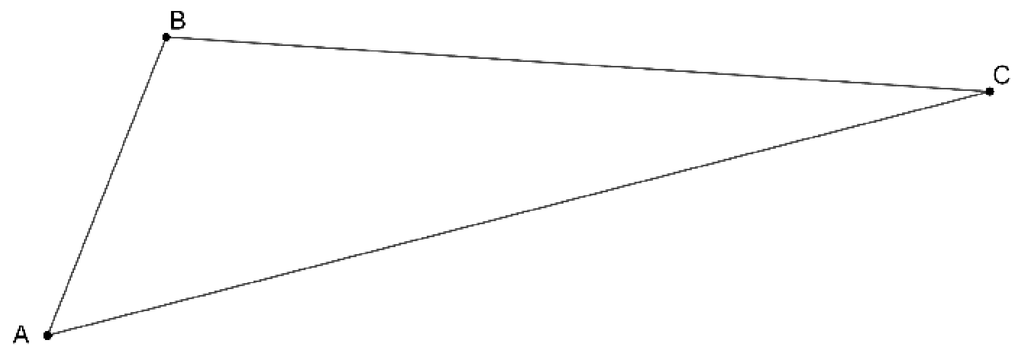
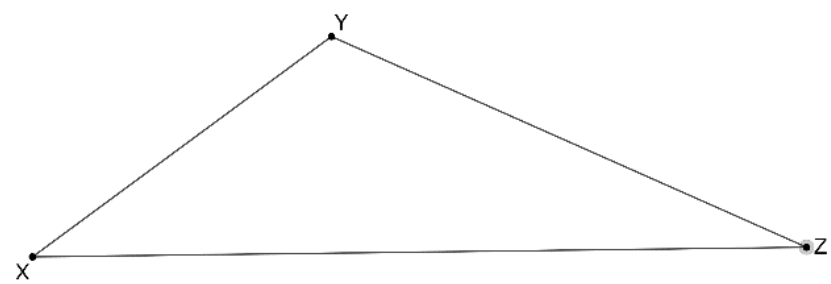
ΔABC is shown, where $AB = 12$, $BC = 20$, and $AC = 26$.



1. List the angles in order from least to greatest.

ΔXYZ is shown. Use the triangle to complete questions 2 through 4.

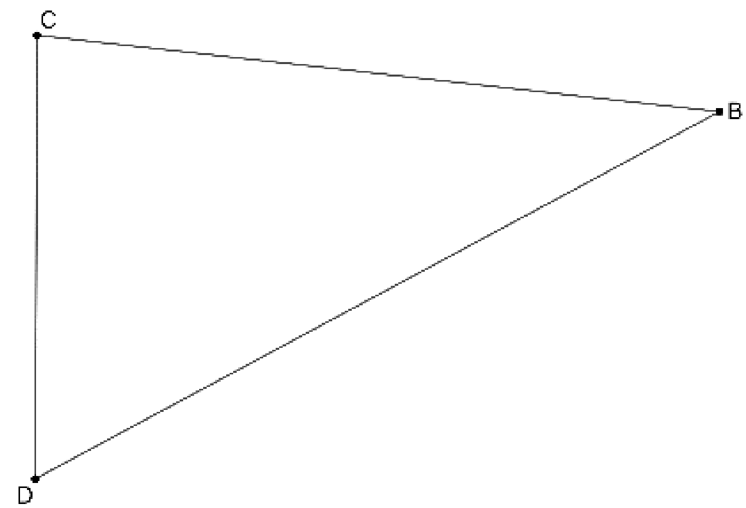


2. Use a protractor to measure each of the angles.

3. List the angles in order from smallest to largest.

4. List the side lengths in order from smallest to largest.

ΔBCD is shown, where $m\angle B = 25°$, $m\angle D = 70°$, and $m\angle C = 85°$.



5. List the side lengths in order from largest to smallest.

For questions 6 through 8, determine whether the given side lengths can form a triangle.

6. $10 \text{ cm}, 11 \text{ cm},$ and 22 cm

7. $3 \text{ in}, 7 \text{ in},$ and 14 in

8. $3 \text{ ft}, 8 \text{ ft},$ and 6 ft

For questions 9 and 10, two sides of a triangle are given. Determine a possible side length for the third side.

9. 10 in and 2 in

10. 5 m and 3 m

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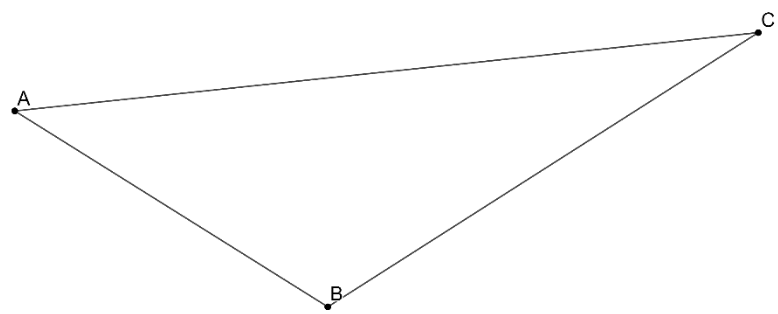
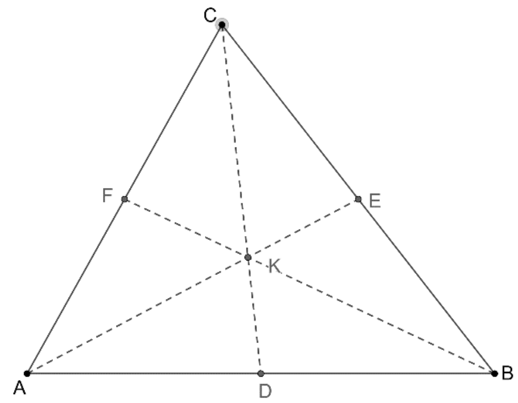
ΔABC is shown. Complete the steps for finding the centroid of the circle. Then, label the centroid Point G.

<table>
<tr><th>Steps</th><th>Construction</th></tr>
<tr><td>1.</td><td rowspan="8">5.
</td></tr>
<tr><td>Draw the median from point R to the opposite vertex, $\angle ACB$.</td></tr>
<tr><td>2.</td></tr>
<tr><td>Draw the median from point T to the opposite vertex, $\angle BAC$.</td></tr>
<tr><td>3.</td></tr>
<tr><td>4.</td></tr>
<tr><td>Draw Point G, the point of concurrency, where the three medians intersect.</td></tr>
</table>

Triangle ABC is shown, where point K is the centroid of the triangle. $FB = 7.5$, $KC = 4.5$, and $KA = 4.8$. Use this information to answer questions 6 through 8.

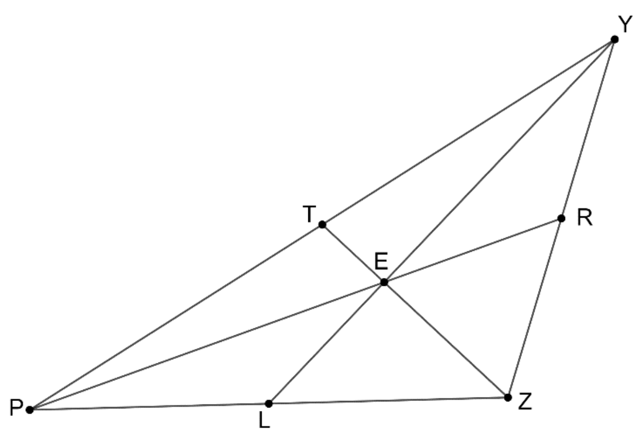


6. Find the length of $\overline{CD}$.

7. Find the length of $\overline{KE}$.

8. Find the length of $\overline{BK}$.

Triangle YPZ is shown, where point E is the centroid of the triangle. $PE = 3x + 8$, and $RE = 2x - 4$. Use this information to answer questions 9 and 10.



9. Find the value of x.

10. Find the length of $\overline{PR}$.

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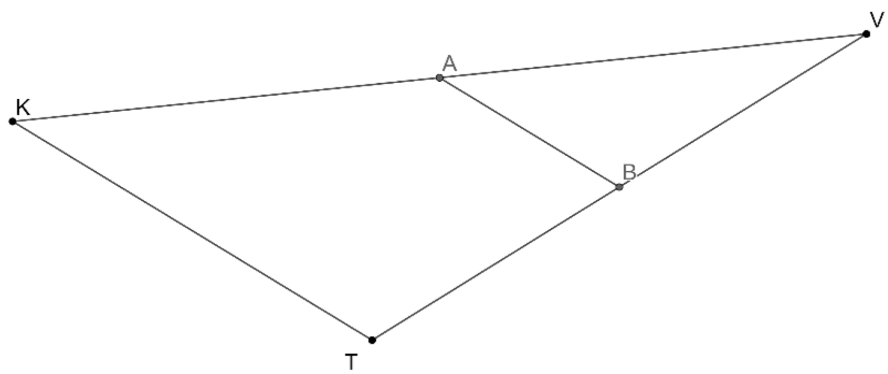
Use this information to answer questions 1 through 5.

Triangle KVT is shown, where $\overline{AB}$ is the midsegment. $KT = 2x + 4$, $AB = \frac{1}{3}x + 4$, $AV = \frac{3}{2}y + 3.5$, $KA = \frac{1}{2}y + 7.5$, $VT = 5z + 3$, and $TB = \frac{3}{4}x + 6$.



1. Find the value of x.

2. Find the value of y.

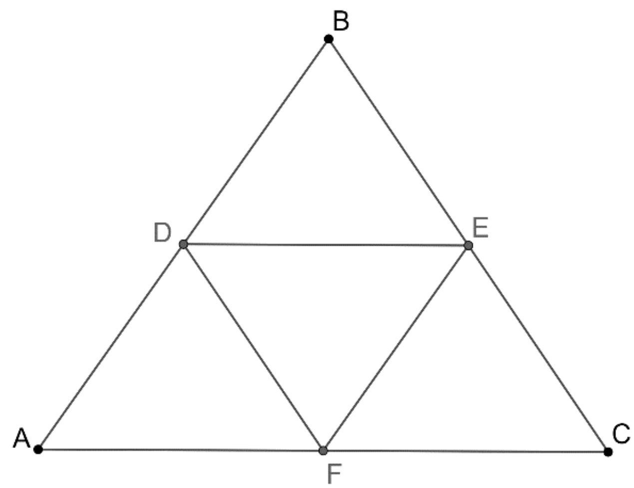
3. Find the value of z.

4. Find the length of the midsegment, $\overline{AB}$.

5. Find the length of $\overline{KV}$.

Use this information to answer questions 6 through 10.

Triangle ABC is shown, where $D, E,$ and F are midpoints. $DE = 2x - 4$, $AC = \frac{1}{2}x + 13$, $DF = \frac{3}{2}x$, $BC = 5x - 12, FE = 3x - 11$, and $AB = \frac{5}{3}x + 4$.



6. Find the value of x.

7. Find the length of $\overline{AB}$.

8. Find the length of $\overline{BC}$.

9. Find the length of $\overline{AC}$.

10. Find the perimeter of triangle DEF.

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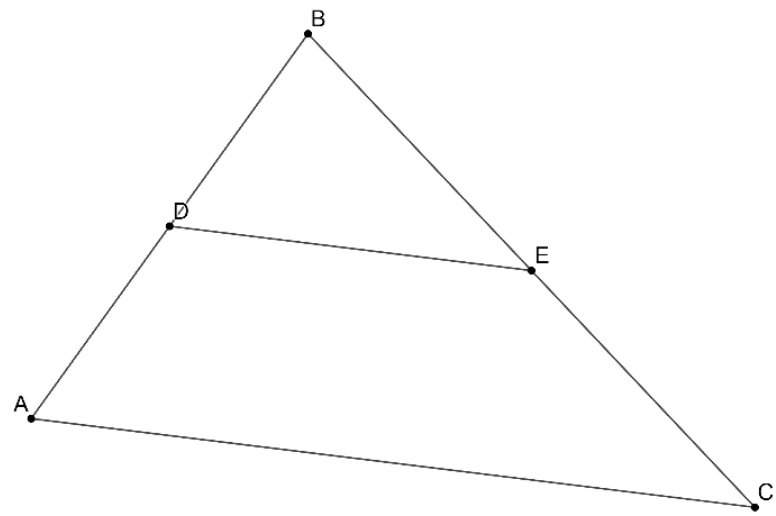
Use this information to answer questions 1 through 5.

Triangle ABC is shown, where $\overline{DE}$ is the midsegment. $AC = 6x + 16$, $DE = 5x - 4$, $AB = 34$, and $BE = 23.8$



1. Find the value of x.

2. Identify the length of $\overline{DE}$.

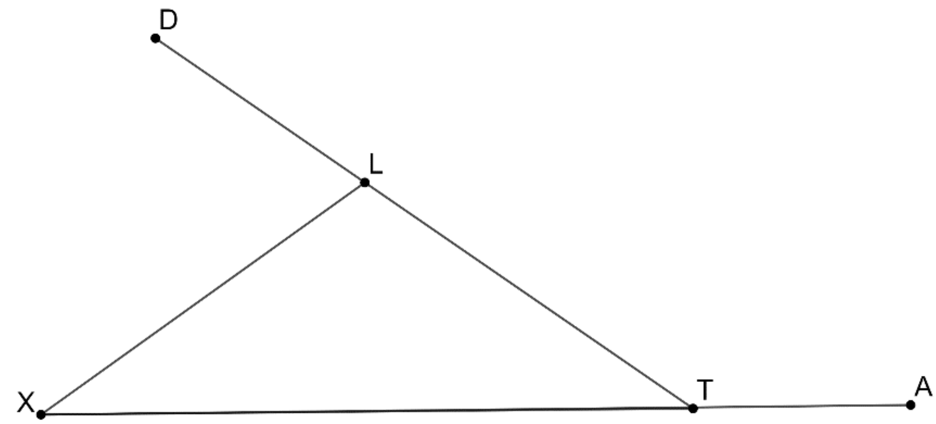
3. Identify the length of $\overline{AC}$.

4. Identify the length of $\overline{AD}$.

5. Calculate the perimeter of triangle BDE.

Use this information to answer questions 6 through 10.

Triangle XTL is shown, where $\angle XLD$ and $\angle LTA$ are exterior angles. $m\angle LXT = m\angle LTX = (7x)°$ and $m\angle XLT = (16x)°$



6. Find the value of x.

7. Find $m\angle XLT$.

8. Find $m\angle LTX$.

9. Find $m\angle DLX$.

10. Find $m\angle LTA$.